

Lab 5 – Solutions

October 27, 2023

1 Thomas algorithm

Consider the *tridiagonal* matrix

$$\mathbb{A} = \begin{bmatrix} a_1 & c_1 & & 0 \\ e_2 & a_2 & \ddots & \\ & \ddots & \ddots & c_{n-1} \\ 0 & & e_n & a_n \end{bmatrix}$$

If the LU decomposition exists, then factors $\mathbb{L}$ and $\mathbb{U}$ are *bidiagonal*, namely

$$\mathbb{L} = \begin{bmatrix} 1 & & & 0 \\ \beta_2 & 1 & & \\ & \ddots & \ddots & \\ 0 & & \beta_n & 1 \end{bmatrix}, \quad \mathbb{U} = \begin{bmatrix} \alpha_1 & c_1 & & 0 \\ & \alpha_2 & \ddots & \\ & & \ddots & c_{n-1} \\ 0 & & & \alpha_n \end{bmatrix}$$

The unknown coefficients α_i and β_i can be determined by imposing the equality $\mathbb{L}\mathbb{U} = \mathbb{A}$. This yields

$$\alpha_1 = a_1, \quad \beta_i = \frac{e_i}{\alpha_{i-1}}, \quad \alpha_i = a_i - \beta_i c_{i-1}, \quad i = 2, \dots, n.$$

Moreover, due to the bidiagonal structure of $\mathbb{L}$ and $\mathbb{U}$, a special version of the forward and backward substitution algorithms can be applied, so that we obtain

Thomas algorithm

$$\begin{aligned} (\mathbb{L}\mathbf{y} = \mathbf{b}) \quad & y_1 = b_1, \quad y_i = b_i - \beta_i y_{i-1}, \quad i = 2, \dots, n \\ (\mathbb{U}\mathbf{x} = \mathbf{y}) \quad & x_n = \frac{y_n}{\alpha_n}, \quad x_i = \frac{y_i - c_i x_{i+1}}{\alpha_i}, \quad i = n-1, \dots, 1 \end{aligned}$$

Exercise 5.1. Consider the tridiagonal matrix $\mathbb{A} \in \mathbb{R}^{10 \times 10}$ defined as

$$\mathbb{A} = \begin{bmatrix} 1 & 11 & & & \\ 102 & 2 & 12 & & \\ & 103 & 3 & 13 & \\ & \cdots & \cdots & \cdots & \\ & & 109 & 9 & 19 \\ & & & 110 & 10 \end{bmatrix}.$$

Then consider the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ such that $\mathbf{x} = \text{ones}(10, 1)$.

- Use the Matlab commands `spdiags` and `sparse` to store the matrix in sparse format, and compare the effect of these commands with respect to the commands `diag` and `full`. Read carefully the documentation of the command `spdiags` using the `help` command.

b. Implement the Thomas algorithm and solve the linear system.

Solution Exercise 5.1.

ex_5_1.m

```
%% Exercise 5.1
clc
clear all
close all

a = [1:10]';
c = [10:19]';
e = [102:111]';
n = length(a);

A_sparse = spdiags([e a c], [-1 0 1], n, n)
% alternative way to create A as a sparse and full matrix
% A = diag(1:10) + diag(102:110,-1) + diag(11:19, 1)
% A_sparse = sparse(A)

xex = ones(n,1);
b = A_sparse*xex

% Compare memory usage of the sparse and full format:
A_full = full(A_sparse)
whos % memory usage of the sparse format is less than the full format

clear Afull

[L,U,x] = thomas(A_sparse,b);
x
```

thomas.m

```
function [L,U,x] = thomas(A,b)
% [L,U] = lu_decomposition(A)
%
% Inputs : A = input tridiagonal matrix in sparse format
%          b = input rhs vector
% Outputs : L = lower triangular matrix
%           U = upper triangular matrix
%           x = solution of the linear system

n = length(b);

c = spdiags(A, 1); % of the form c = [0, c_1, c_2, ... c_{n-1}]
a = spdiags(A, 0); % of the form a = [a_1, a_2, ... a_n]
e = spdiags(A,-1); % of the form e = [e_2, e_3, ... e_n, 0]

y = zeros(n,1);
x = zeros(n,1);
alpha = zeros(n,1); % of the form alpha = [alpha_1, alpha_2, ... alpha_n]
beta = zeros(n,1); % of the form beta = [beta_2, beta_3, ... beta_n, 0]
eyediag = ones(n,1);

%% LU factorization %%
alpha(1) = a(1);
for i=2:n
    beta(i-1) = e(i-1)/alpha(i-1);
    alpha(i) = a(i) - beta(i-1)*c(i);
end
L = spdiags([beta eyediag], [-1 0], n, n);
U = spdiags([alpha c], [0 1], n, n);

%% Forward substitution %%
y(1)=b(1);
for i=2:n
    y(i) = b(i) - beta(i-1)*y(i-1);
```

```

end

%% Backward substitution %%
x(n) = y(n) / alpha(n);
for i = n-1:-1:1
    x(i) = (y(i) - c(i+1)*x(i+1))/alpha(i);
end

end

```

2 Condition number

Definition 5.1. Let V be a vector space associated with the field $K = \mathbb{R}$ or $\mathbb{C}$. The map $\|\cdot\|$ from V into $\mathbb{R}$ is a **norm** in V if

1. $\|\mathbf{v}\| \geq 0 \quad \forall \mathbf{v} \in V$ and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$;
2. $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\| \quad \forall \alpha \in K, \forall \mathbf{v} \in V$;
3. $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\| \quad \forall \mathbf{v}, \mathbf{w} \in V$.

In $V = \mathbb{R}^n$ we can define the **p -norm**

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \quad 1 \leq p < \infty;$$

when $p = 2$ we recover the **Euclidean norm**. For $p = \infty$ we define the so called **infinity norm**

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

Definition 5.2. Consider $\mathbb{A} \in \mathbb{R}^{m \times n}$. The mapping $\|\cdot\| : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is a **matrix norm** if it satisfies the same properties as for vector norms. Moreover, a matrix norm is said **submultiplicative** if it satisfies

$$4. \|\mathbb{A}\mathbb{B}\| \leq \|\mathbb{A}\| \|\mathbb{B}\| \quad \forall \mathbb{A}, \mathbb{B} \in \mathbb{R}^{m \times n}.$$

The matrix 1-norm and matrix ∞ -norm are defined by

$$\|\mathbb{A}\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}| \quad \text{and} \quad \|\mathbb{A}\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$$

namely by the so called **column sum** and **row sum norm**, respectively. The 2-norm is called **spectral norm** and is defined by $\|\mathbb{A}\|_2 = \sqrt{\lambda_{\max}(\mathbb{A}^T \mathbb{A})}$.

Definition 5.3. The **condition number** with respect to generic norm p is defined as

$$\mathcal{K}_p(\mathbb{A}) = \|\mathbb{A}\|_p \|\mathbb{A}^{-1}\|_p$$

Given a system $\mathbb{A}\mathbf{x} = \mathbf{b}$ we consider the problem where the right-hand side term is **perturbed** by vector $\delta \mathbf{b}$

$$\mathbb{A}(\mathbf{x} + \delta \mathbf{x}) = \mathbf{b} + \delta \mathbf{b}$$

$\mathcal{K}_p(\mathbb{A})$ is a **measure of the sensitivity** of the solution $\mathbf{x}$ to changes in $\mathbf{b}$ and it holds

$$\frac{\|\delta \mathbf{x}\|_p}{\|\mathbf{x}\|_p} \leq \mathcal{K}_p(\mathbb{A}) \frac{\|\delta \mathbf{b}\|_p}{\|\mathbf{b}\|_p}$$

- $\mathcal{K}_p(\mathbb{A}) = 10^k$: you **may lose** up to k digits of accuracy in the solution
- **MATLAB**: `cond(A,p)` (`cond(A)`, $p = 2$ by default)

Exercise 5.2. Consider the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbb{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \varepsilon \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

where $0 < \varepsilon \ll 1$.

- Compute $\mathcal{K}_p(\mathbb{A})$ for $p = 1, 2, \infty$ without using MATLAB. Check the results in MATLAB for some values of ε ($1e-6, 1e-8$).
- For $\varepsilon = 1e-6$, and $\alpha = 10^{-12}, 10^{-6}$, consider the perturbation term $\delta\mathbf{b} = [0 \ 0 \ \alpha]^T$, and compute the associated perturbation on the solution $\mathbf{x}$.
- Consider now $\delta\mathbf{b} = [\alpha \ 0 \ 0]^T$ and compute the associated perturbation on the solution $\mathbf{x}$

Solution Exercise 5.2.

ex_5_2.m

```
%% Exercise 5.2
clc
clear all
close all

%% Compute Kp (A) for p = 1, 2, without using MATLAB. Check the results
% in MATLAB for some values of (1e -6, 1e -8).
% 2-norm
% norm(A,2) = sqrt(eig_max(A'A))
% In this case A is diagonal A'A = A^2
% norm(A,2)=sqrt(eig_max(A^2)) = sqrt((eig_max(A))^2) = eig_max(A) = 1.
% For the same reasons, norm(A^-1,2) = 1/epsilon
% => K2 = norm(A,2) * norm(A^-1,2) = 1/epsilon.

% 1-norm / infinity-norm
% Since A is symmetric, the two norms coincide. In particular:
% norm(A,1) = max_j(sum_i(abs(aij))) = 1
% norm(A^-1,1) = max_j(sum_i(abs(aij^-1))) = 1/epsilon
% K1 = Kinf = 1/epsilon

A = eye(3);
epsilon = 1e-6;
A(3,3) = epsilon;

eigenvalues=eig(A)

K2=cond(A,2)

K1=cond(A,1)

Kinf=cond(A,'inf')

%% For epsilon = 1e6, and alpha =( 1012 , 106) , consider the perturbation term
% delta_b = [0 0 alpha] , and compute the associated perturbation on the solution x

% deltax = A^-1 deltab = [0 0 alpha/epsilon]'
% c)
% deltax = A^-1 deltab = [alpha 0 0]'

% Both cases verify the inequality: norm(deltax)/norm(x) <= K
% norm(deltab)/norm(b).
% In particular, in the case b) we have an equality
% and in the case c) we have a strict inequality.

b = [1 0 0]';
x_ex = [1 0 0]';

for (alpha = [1e-6 1e-12])
    deltab_1 = [0 0 alpha]';
```

```

    max_deltax_inf=Kinf*norm(deltab_1, inf)/norm(b, inf)% K(A)_p ||deltab||_p/||b||_p, p=inf
    deltax = A \ ( deltab_1) % delta x
    norm_deltaxx=norm(deltax, inf) / norm(x_ex, inf) % ||deltax||_p/||x||_p, p=inf

    %check
    % ||deltax||_p/||x||_p<=K(A)_p ||deltab||_p/||b||_p, p=inf
    norm_deltaxx<=max_deltax_inf
end

%% Consider now deltab = [alpha 0 0]' and compute the associated perturbation on the solution
x
for (alpha = [1e-6 1e-12])
    deltab_2 = [alpha 0 0]';
    max_deltax_inf2=Kinf*norm(deltab_2, inf)/norm(b, inf)% K(A)_p ||deltab||_p/||b||_p, p=inf
    deltax2 = A \ ( deltab_2) % delta x
    norm_deltaxx2=norm(deltax, inf) / norm(x_ex, inf) % ||deltax||_p/||x||_p, p=inf

    %check
    % ||deltax||_p/||x||_p<=K(A)_p ||deltab||_p/||b||_p, p=inf
    norm_deltaxx2<=max_deltax_inf2
end

```

Exercise 5.3. Consider the Hilbert matrix $\mathbb{H}_n$ of order n

$$\mathbb{H}_n = \begin{bmatrix} 1 & 1/2 & \cdots & 1/n \\ 1/2 & 1/3 & \cdots & 1/(n+1) \\ \vdots & \vdots & \ddots & \vdots \\ 1/n & 1/(n+1) & \cdots & 1/(2n-1) \end{bmatrix} \in \mathbb{R}^{n \times n}, \quad n = 10, 20, 40.$$

- Estimate $\mathcal{K}_2(\mathbb{H}_n)$ using the technique of perturbing a linear system with $\mathbb{H}_n$ as coefficient matrix. Consider the right hand side term b such that $x = \text{ones}(n, 1)$.
- Compute $\mathcal{K}_2(\mathbb{H}_n)$ using MATLAB command `cond`.
- What can you say about the **accuracy of the solution** of the generic linear system $\mathbb{H}_n x = b$? How large can you take n before the relative error is 1 (i.e. there are no significant digits in the solutions)?

Solution Exercise 5.3.

ex_5_3.m

```

%% Homework 5.1
clc
clear all
close all

% We can invert the relation
% norm(deltax)/norm(x) <= norm(deltab)/norm(b)
% to get an estimate for Kp(A):
% Kp(A) >= norm(deltax)/norm(x) * norm(b)/norm(deltab)

for (n = [10 20 40])
    pert_mag = 1e-6; % Perturbation magnitude
    n_trials = 100;
    H = hilb(n);
    x = ones(n, 1);
    b = H*x;
    KH_est = 0;
    for (ii = 1:n_trials)
        deltab = randn(n, 1) * pert_mag;
        warning('off', 'all')
        deltax = H \ deltab;
        KH_est_ii = norm(deltax)/norm(x) * norm(b)/norm(deltab);
        KH_est = max(KH_est_ii, KH_est);
    end
    KH_est
end

```

```

cond( hilb(10) )
cond( hilb(20) )
cond( hilb(40) )

% The estimates obtained with the perturbing technique are lower. This is expected, since
% the condition number represents the maximum
% possible amplification of the relative error on data. With the first
% approach we might not have considered the worst condition,
% hence the estimate is lower than the result with cond command.

n = [1:20];

for (ii = n)
    H = hilb(ii);
    x_ex = ones(ii, 1);
    b = H*x_ex;
    x = H\b;
    KH(ii) = cond(H);
    err(ii) = norm(x - x_ex) / norm(x);
end

figure
semilogy(n, KH, 'xb-', 'LineWidth',2, 'MarkerSize',8)
hold on
semilogy(n, err, 'xr-', 'LineWidth',2, 'MarkerSize',8)
semilogy(n, ones(n(end),1), 'k--', 'LineWidth',2, 'MarkerSize',8)
set(gca, 'FontSize',16)
set(gca, 'LineWidth', 1.5)
axis([0 n(end)+1 1e-17 1e20])
xlabel('n')
ylabel('K_2(H_n) and err')
legend('K_2(H_n)', 'relative error', 'Location','SouthEast')

% For n > 12 the relative error is 1. This means that the solution is completely meaningless.

```

3 Iterative methods

According to a generic iterative **method**, the solution $\mathbf{x}$ to the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ is approximated by

$$\mathbf{x}^{(k+1)} = \mathbb{B}\mathbf{x}^{(k)} + \mathbf{g}, \quad \mathbb{B} : \text{iteration matrix}$$

such that

$$\mathbf{x} = \mathbb{B}\mathbf{x} + \mathbf{g} \quad \text{and} \quad \mathbf{x} = \lim_{k \rightarrow \infty} \mathbf{x}^{(k)}$$

(consistency)

(convergence)

Usually $\mathbf{x}$ is not reached in a finite number of iterations, so a **stopping criterion** is applied

- **increment** $\|\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)}\| \leq \varepsilon$
- **residual** $\|\mathbf{b} - \mathbb{A}\mathbf{x}^{(k)}\| \leq \varepsilon.$

If the method is consistent

$$\mathbf{x}^{(k+1)} = \mathbb{B}\mathbf{x}^{(k)} + \mathbf{g} \text{ converges} \iff \rho(\mathbb{B}) < 1$$

$R(\mathbb{B}) = -\log \rho(\mathbb{B})$ **asymptotic convergence rate**.

The square matrix $\mathbb{A}$ can be written as $\mathbb{A} = -\mathbb{E} + \mathbb{D} + (\mathbb{A} - \mathbb{D} + \mathbb{E})$ where

- $-\mathbb{E}$: lower triangular part starting from the first subdiagonal (MATLAB: $-E = \text{tril}(A, -1)$)
- $-\mathbb{F}$: upper triangular part starting from the first superdiagonal (MATLAB: $-F = \text{triu}(A, 1)$)
- $\mathbb{D}$: diagonal matrix extracted from $\mathbb{A}$ (MATLAB: $\mathbb{D} = \text{diag}(\mathbb{A})$)

Jacobi method

$$\mathbb{B}_J = \mathbb{I} - (\mathbb{D})^{-1}\mathbb{A} \quad \mathbf{g}_J = \mathbb{D}^{-1}\mathbf{b}$$

Gauss-Seidel method

$$\mathbb{B}_{GS} = (\mathbb{D} - \mathbb{E})^{-1}\mathbf{F} \quad \mathbf{g}_{GS} = (\mathbb{D} - \mathbb{E})^{-1}\mathbf{b}$$

Convergence Jacobi

$\mathbb{A}$ is strictly diagonally dominant by rows or by columns $\implies$ Jacobi and Gauss-Seidel methods converge for any initial guess $\mathbf{x}^{(0)}$

Convergence Gauss-Seidel

$\mathbb{A}$ is symmetric and positive definite $\implies$ Gauss-Seidel method converges for any initial guess $\mathbf{x}^{(0)}$

Exercise 5.4. Consider the tridiagonal matrix $\mathbb{A} \in \mathbb{R}^{10 \times 10}$ defined as

$$\mathbb{A} = \begin{bmatrix} 3 & -2 & & & \\ -1 & 3 & -2 & & \\ & \dots & \dots & \dots & \\ & & -1 & 3 & -2 \\ & & & -1 & 3 \end{bmatrix}.$$

Consider also the linear system $\mathbb{A}\mathbf{x} = \mathbf{b}$ such that $\mathbf{x} = \text{ones}(10, 1)$.

- Implement the Jacobi and Gauss-Seidel methods and solve the linear system.
- Do both the Jacobi and the Gauss-Seidel methods converge? Numerically confirm the relation between their spectral radii.
- Apply both methods, with $\mathbf{x}^{(0)} = \mathbf{0}$ and with a tolerance of 10^{-12} , and compare the required number of iterations.

Solution Exercise 5.4.

ex_5_4.m

```
%% Homework 5.5
clc
clear all
close all

n = 10;
A = 3*eye(n) - 2*diag(ones(n-1, 1), 1) - diag(ones(n-1, 1), -1);
b = A*ones(n, 1);

% Iteration matrices.
D = diag(diag(A));
E = -tril(A, -1);
F = -triu(A, 1);

Bj = eye(size(A)) - D\A; % Jacobi method
Bgs = (D-E) \ F; % Gauss-Seidel method

% Compute the spectral radii of both matrices to determine whether the
% methods are convergent or not.

rhoBj = max(abs(eig(Bj)))
rhoBgs = max(abs(eig(Bgs)))

% It can be noted that both spectral radii are less than 1, so that
% both method are convergent, and that the spectral radius of the Gauss-Seidel
% method is the square of the spectral radius of the Jacobi method.

% Iteration vector
gj = D \ b;
```

```

ggs = (D-E) \ b;

x0 = zeros(n, 1);
tol = 1e-12;
maxit = 1000;

% Jacobi method
[xj, iterj, incrgj] = stationary_method(Bj, gj, x0, tol, maxit)
% or you can use the jacobi function
%[xj, iterj]=jacobi(A,b,x0,tol, maxit);
% Gauss-Seidel method
[xgs, itergs, incrgs] = stationary_method(Bgs, ggs, x0, tol, maxit)
% or you can use the gs function
%[xgs, itergs]=gs(A,b,x0,tol, maxit);
% The number of iterations of the Gauss-Seidel method is approximately half
% of the ones of the Jacobi method.

```

Exercise 5.5 (Exam Sep 7, 2023). Consider the matrices $\mathbb{A}$ and $\mathbb{B}$ defined by the following MATLAB commands:

$$\begin{cases} n=100; \\ \mathbb{A}=-19*\text{diag}(\text{ones}(n-2,1),-2) - 7/3*\text{diag}(\text{ones}(n,1)) + \\ +3*\text{diag}(\text{ones}(n-1,1),1) + 3*\text{diag}(\text{ones}(n-1,1),-1)+ \\ +57/3*\text{diag}(\text{ones}(n-2,1),-2); \\ \mathbb{C}=-7*\text{diag}(\text{ones}(n,1)) + 2*\text{diag}(\text{ones}(n-2,1),-2) \\ +2*\text{diag}(\text{ones}(n-2,1),2); \\ \mathbb{B}=\mathbb{C}'*\mathbb{C}; \end{cases}$$

- Solve Numerically the linear systems $\mathbb{A}\mathbf{x} = \mathbf{a}$, $\mathbb{B}\mathbf{x} = \mathbf{b}$ with $\mathbf{a} = \mathbb{A} * \text{ones}(n,1)$, $\mathbf{b} = \mathbb{B} * \text{ones}(n,1)$. Resort to the cheapest method in terms of computational cost.
- Compute the error associated with the two approximations computed at the previous point, in terms of the infinity norm.

Solution Exercise 5.5.

ex_5_5.m

```

%% Exam Sep 7, 2023
clc; close all; clear

n=100;
A=-19*diag(ones(n-2,1),-2) - 7/3*diag(ones(n,1)) +...
+3*diag(ones(n-1,1),1) + 3*diag(ones(n-1,1),-1)+...
+57/3*diag(ones(n-2,1),-2);
% Let's look at the matrix
figure()
spy(A)
xex=ones(n,1);
a=A*xex;
% It's tridiagonal: cheapest method is the thomas alghoritm
[L,U,xa] = thomas(A,a);
xa
keyboard
C=-7*diag(ones(n,1))+ 2*diag(ones(n-2,1),-2) +2*diag(ones(n-2,1),2);
B=C'*C;
% Let's look at the matrix
figure()
spy(B)
b=B*xex;

% The matrix is penta-diagonal and simmetric and positive definite
%%
any(eig(B)<=0)
% we can use the cholesky factorization

H=chol(B);
y = H'\b;

```

```
xb = H\y
```

```
%%
err1=norm(xa-xex,inf)/norm(xex, inf)
err2=norm(xb-xex,inf)/norm(xex, inf)
```

Exercise 5.6 (Exam June 17, 2024). Consider the family of matrices $\mathbb{A}_n(a) \in \mathbb{R}^{n \times n}$ defined by:

$$\mathbb{A}_n(a) = \begin{bmatrix} 1 & -11 & 0 & 0 & \dots \\ -11 & 1 & -12 & 0 & \dots \\ 0 & -12 & 1 & -13 & \dots \\ 0 & 0 & -13 & \ddots & \ddots \\ \vdots & \vdots & \vdots & \ddots & \ddots \end{bmatrix} + a\mathbb{I}_n.$$

where $\mathbb{I}_n$ is the identity matrix of order n .

Exam June 17, 2024

- Resorting to the MATLAB commands `diag` and `eye`, store the matrix $\mathbb{A}_7(32)$. Report the commands used.
- Resorting to the MATLAB command `lu`, compute the LU factorization of $\mathbb{A}_7(32)$ and answer to the following questions (use the commands `nnz` and `spy`), providing justifications:

- Was the pivoting performed?
- did fill-in occur?

Include the MATLAB commands used in your answer.

- Set $\mathbf{x}_{ex} = [1, 1, 1, \dots]^T \in \mathbb{R}^n$ and consider the linear system $\mathbb{A}_n(a)\mathbf{x} = \mathbf{b}$ with $\mathbf{b} = \mathbb{A}_n(a)\mathbf{x}_{ex}$. Use the provided `jacobi.m` function to solve the linear system for $n = 7$ and $a = 1$ and compute the error $\eta_7(1) = \|\mathbf{x}_{ex} - \mathbf{x}\|_2$. Set `tol=1e-6` and `nmax=100`. Choose the initial guess $\mathbf{x}^{(0)} = \mathbf{0}$. Comments the results and report the MATLAB commands.
- Use the provided `jacobi.m` function to solve the linear system for $n = 7$ and $a = 32$ and compute the error $\eta_7(32) = \|\mathbf{x}_{ex} - \mathbf{x}\|_2$. Set `tol=1e-6` and `nmax=100`. Choose the initial guess $\mathbf{x}^{(0)} = \mathbf{0}$. Compare the results with the previous case and discuss them considering the theoretical expectations.

Solution Exercise 5.6.

ex_5_6.m

```
%% 2a
% The MATLAB commands are:
n = 7;
a = 32;
B = eye (n) - diag (10+[1:n-1], -1) - diag (10+[1:n-1], +1);
% The result is:
A = B + a * eye (n)
```

```
%% 2b
% LU factorization of A
[L, U, P] = lu (A);
% Patterns of A, L, U, and P
figure (1)
spy (A)
figure (2)
spy (L, 'x')
hold on
spy (U, 'o')
figure (3)
spy (P)
% Number of nonzeros in A, L, and U
nnz (A);
nnz (L);
```

```

nnz(U);

% From the sparsity pattern and from the number of nonzeros in L, U, A, and P, it is clear
that:
% -The pivoting was not performed, indeed, the sparsity pattern of P is the identity matrix.
% - The fill-in does not occur, indeed, the sparsity pattern of L and U is the same of A.
% - The number of nonzeros in L and U is the same of A plus 7 : nnz(L)+nnz(U) = nnz(A)+7.
% This is because the seven diagonal elements are counted twice, once in L and once in U,
because they are shared by the two matrices.

%% 2c
n = 7;
a = 1;
B = eye(n) - diag(10+[1:n-1], -1) - diag(10+[1:n-1], +1);
A = B + a * eye(n);
xex = ones(n, 1);
b = A * xex;
[y, k] = jacobi(A, b, b*0, 1e-6, 100)
norm(y - xex, 2)
% The method does not converge because the spectral radius of the iteration matrix is greater
than 1.
Bj=eye(n)-diag(diag(A))\A;
rhobj=max(abs(eig(Bj)))

%2d
n = 7;
a = 32;
B = eye(n) - diag(10+[1:n-1], -1) - diag(10+[1:n-1], +1);
A = B + a * eye(n);
xex = ones(n, 1);
b = A * xex;
[y, k] = jacobi(A, b, b*0, 1e-6, 100)
norm(y - xex, 2)
Bj=eye(n)-diag(diag(A))\A;
rhobj=max(abs(eig(Bj)))
% The method performed 54 iterations.
% The error in the 2-norm of the solution, in this case, is of order 10^-6.
% This is consistent with the theoretical result, which guarantees the convergence of the
Jacobi method for strictly diagonally dominant matrices by rows.

```

Exercise 5.7 (Exam July 21, 2021). Consider the matrix $\mathbb{A}$:

$$\mathbb{A}_n(a) = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & 1 \\ 0 & -1 & 4 \end{bmatrix} + a\mathbb{I}_n.$$

Set $\mathbf{b}$ such that $\mathbf{x}_{ex} = [1/2, 1/3, 1/4]^T$. With the aim of solving the linear system $\mathbb{A}_n(a)\mathbf{x} = \mathbf{b}$, with an iterative method, set $\mathbf{x} = [0, 0, 0]^T$ and $N_{max} = 100$.

Exam July 21, 2021

- Using the `jacobi.m` function solve the linear system with the Jacobi iterative method with the following tolerances: `tol = 10-2, 10-3, 10-4`. Report the obtained solutions and the number of iterations required for each case and the MATLAB instructions.
- Comment the results obtained in the previous point.
- By using the function `gs.m` solve the linear system with the Gauss-Seidel iterative method with tolerance `tol = 10-4`. Report the obtained solution and the number of iterations required. Include the MATLAB instructions.
- Regarding point **a** with `tol = 10-4` and point **c**, explain why the two methods converge and compare the convergence of the two methods, justifying their behavior in light of the theory.
- Referring to the iteration matrices constructed in `jacobi.m` and `gs.m`, calculate the spectral radius of the two methods for the proposed exercise and compare the results in light of the findings obtained in point **d**.

Solution Exercise 5.7.

ex_5_6.m

```

%% 2a
% The MATLAB commands are:
n = 7;
a = 32;
B = eye (n) - diag (10+[1:n-1], -1) - diag (10+[1:n-1], +1);
% The result is:
A = B + a * eye (n)

%% 2b
% LU factorization of A
[L, U, P] = lu (A);
% Patterns of A, L, U, and P
figure (1)
spy (A)
figure (2)
spy (L, 'x')
hold on
spy (U, 'o')
figure (3)
spy (P)
% Number of nonzeros in A, L, and U
nnz(A);
nnz(L);
nnz(U);

% Fromt the sparsity pattern and from the number of nonzeros in L, U, A, and P, it is clear
that:
% -The pivoting was not performed, indeed, the sparsity pattern of P is the identity matrix.
% - The fill-in does not occur, indeed, the sparity pattern of L and U is the same of A.
% - The number of nonzeros in L and U is the same of A plus 7 : nnz(L)+nnz(U) = nnz(A)+7.
% This is because the seven diagonal elements are counted twice, once in L and once in U,
because they are shared by the two matrices.

%% 2c
n = 7;
a = 1;
B = eye (n) - diag (10+[1:n-1], -1) - diag (10+[1:n-1], +1);
A = B + a * eye (n);
xex = ones(n, 1);
b = A * xex;
[y, k] = jacobi (A, b, b*0, 1e-6, 100)
norm (y - xex, 2)
% The method does not converge because the spectral radius of the iteration matrix is greater
than 1.
Bj=eye(n)-diag(diag(A))\A;
rhobj=max(abs(eig(Bj)))

%2d
n = 7;
a = 32;
B = eye (n) - diag (10+[1:n-1], -1) - diag (10+[1:n-1], +1);
A = B + a * eye (n);
xex = ones(n, 1);
b = A * xex;
[y, k] = jacobi (A, b, b*0, 1e-6, 100)
norm (y - xex, 2)
Bj=eye(n)-diag(diag(A))\A;
rhobj=max(abs(eig(Bj)))
% The method performed 54 iterations.
% The error in the 2-norm of the solution, in this case, is of order 10^-6.
% This is consistent with the theoretical result, which guarantees the convergence of the
Jacobi method for strictly diagonally dominant matrices by rows.

```